

Regression and Multivariate Data Analysis

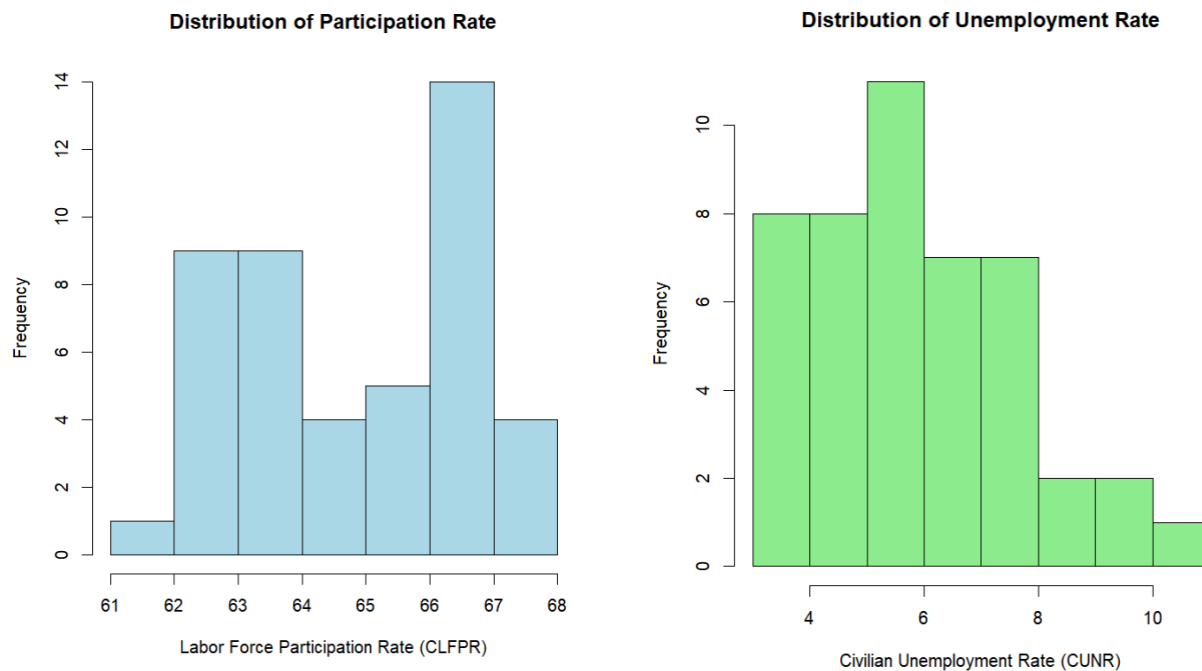
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Unemployment Data Analysis

In this paper I will analyze the relationship between unemployment rates and labor force participation rate in the United States from 1980 to 2025. To be exact, I will be looking at whether the civilian unemployment rate (CUNR) is a reliable predictor for the civilian labor force participation rate (CLFPR). The data used was taken from the Current Population Survey (CPS), which allows us to test two economic theories: the discouraged worker effect, which suggests that during tough economic times, people give up looking for jobs and drop out of the labor force. The second theory is the added worker effect, which suggests that when the economy is struggling, new workers like stay-at-home parents join the labor force to make up for lost family income. First, I will look at each variable one at a time to describe their basic patterns. After that, I will do a deeper analysis to see how the two variables work together during two different time periods. The programming language, R, was used for the analysis.

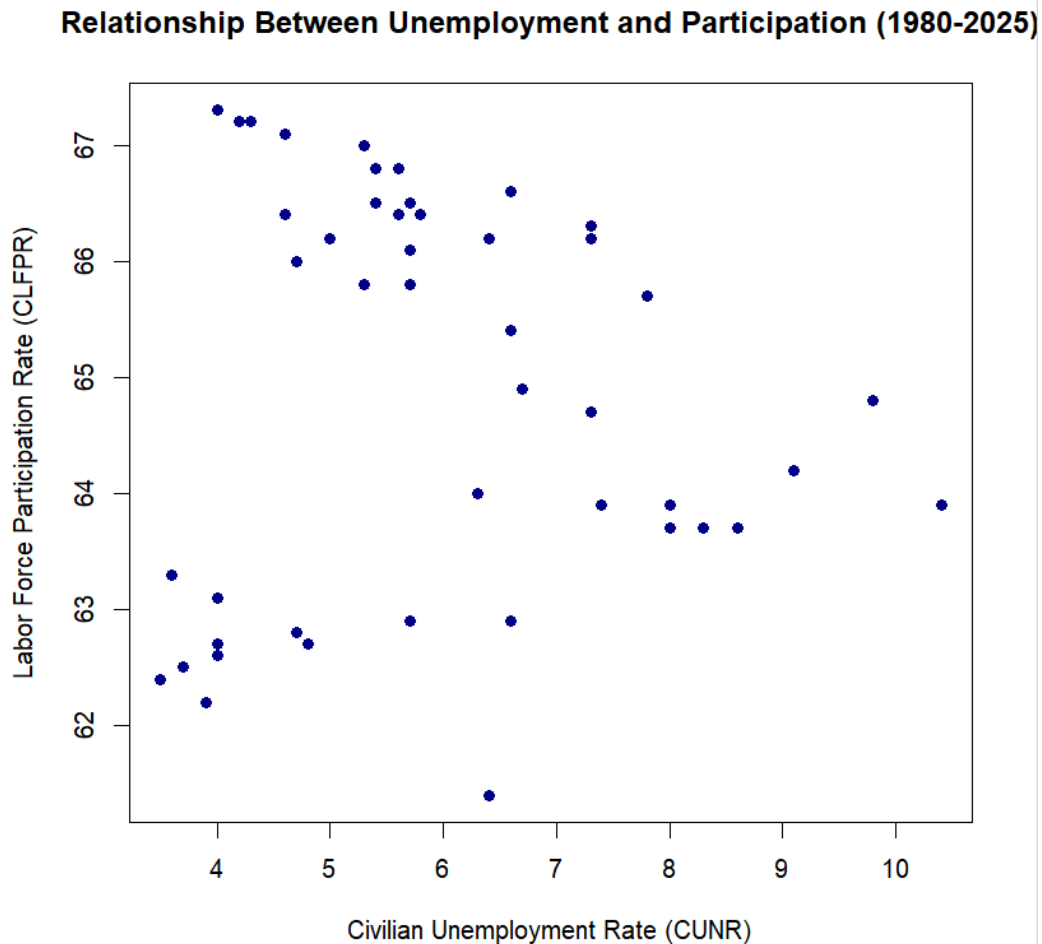
Performing a univariate analysis allows us to see the statistical characteristics and distribution shapes for each variable. The distribution of the labor force participation rate (CLFPR) and the civilian unemployment rate (CUNR) shows how these economic indicators behaved over the last 44 years. The participation rate has an average of about 64.89% and a median of 65.15%. This variable has a standard deviation of 1.72 and a range of 5.90%, spanning from a minimum of 61.40% to a maximum of 67.30%. Its distribution is fairly even, though it leans slightly to the left. On the other hand, the unemployment rate has an average of 5.95%, a lower median of 5.70%, and a standard deviation of 1.71. The unemployment data has a range of

6.90%, with a minimum value of 3.50% and a maximum of 10.40%. This distribution is skewed to the right because a few years had very high unemployment that pulled the average up. Both variables have a similar amount of spread in their data. While there are no extreme univariate outliers that fall more than three standard deviations from the mean, the high-unemployment years represent the most unusual points in this dataset.



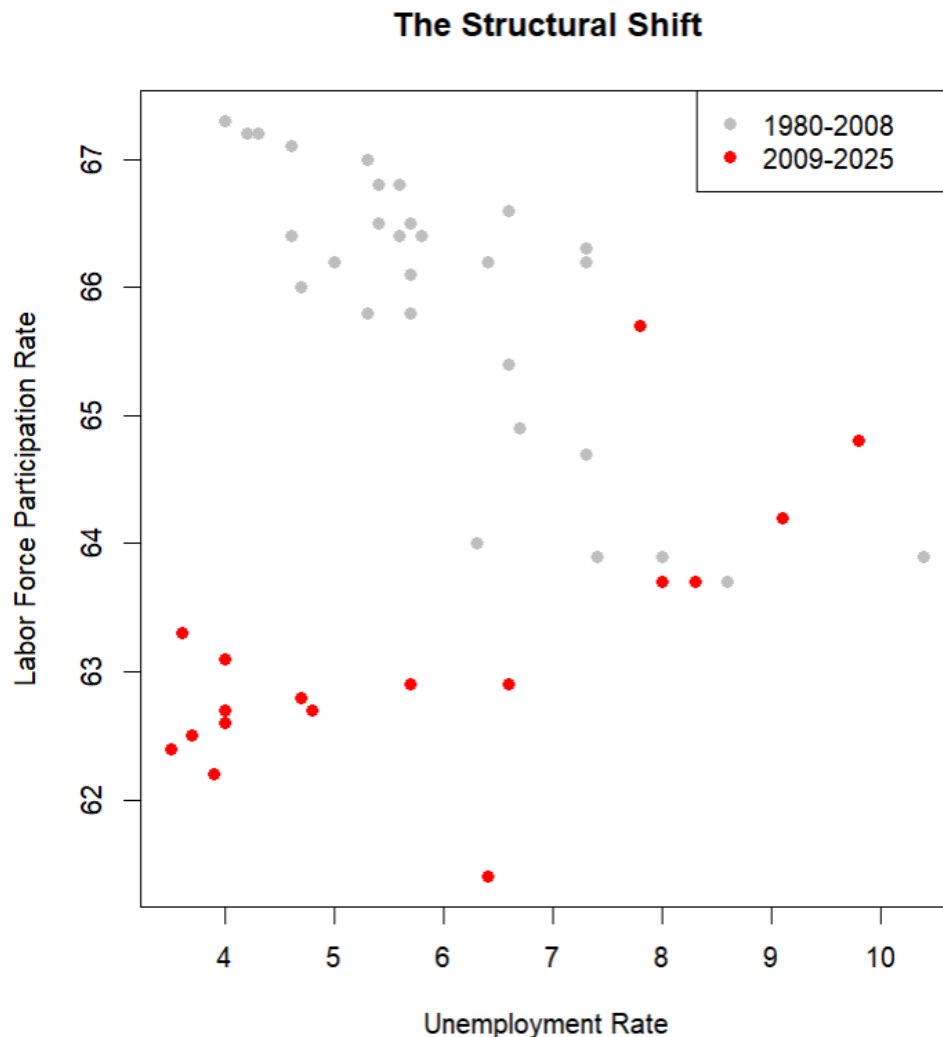
A bivariate analysis of the full dataset shows that there is no statistically significant relationship between the civilian unemployment rate and the labor force participation rate over the entire 1980–2025 period. Based on the scatterplot below, the data points are spread out with no clear pattern, and the correlation is almost zero at -0.02. This means that over the long term, knowing the unemployment rate does not help us predict the participation rate. Because this

relationship is not statistically significant, a single regression model for the whole 44-year span is not useful.



By performing a second look at the bivariate relationship across specific time periods, we can see how the link between unemployment and participation changed over time. When the data is divided into two eras, 1980 to 2008 and 2009 to 2025, the visual evidence suggests a structural shift in the economy. In the first era (1980–2008), there was a strong negative correlation of -0.79, indicating that as unemployment rose, participation tended to fall. However, in the second era (2009–2025), the relationship switched to a strong positive correlation of 0.68,

meaning the two variables began to move in the same direction. Both are statistically significant with a p-value less than 0.01. These separate correlation analyses provide a clear measurement for what we see in the graphs, confirming that the relationship shifted during the 44-year period. Since the relationship appears linear for both eras, a regression is appropriate.



Conducting two separate regression analyses allows us to predict the labor force participation rate based on the unemployment rate for each specific historical era. Because the initial bivariate analysis showed that the relationship between these variables changed over time,

running separate models provides a more accurate picture of the economic trends. Both models are statistically significant because their p-values are less than 0.01. For the first era (1980–2008), the model produced an R^2 value of 0.63 and a slope of -0.62, indicating that the unemployment rate explains about 63% of the variation and that participation dropped as unemployment rose. This creates a regression equation of $CLFPR = 69.66 - 0.62(CUNR)$. In contrast, the second era (2009–2025) resulted in an R^2 of 0.46 and a positive slope of 0.33, suggesting that for every 1% increase in unemployment, participation was predicted to rise by about 0.33%. This creates a regression equation that is $CLFPR = 61.28 + 0.33(CUNR)$. Both models were statistically significant, confirming that unemployment is a reliable predictor within these eras, though the intercept was not interpreted as an unemployment rate of 0% is not a realistic economic scenario.

Next, I will conduct a residual analysis to help us find any bivariate outliers and influential data points. By looking at "standardized residuals," we can find outliers, which are years where the actual participation rate was much higher or lower than the model expected. We use Cook's Distance to identify "influential" years that disproportionately skewed the model's trend. In the first era (1980–2008), 1980 was an outlier and 1983 was a very influential year, while in the second era (2009–2025), 2009 and 2021 stood out as outliers, but only 2009 as an influential data point. These specific years usually line up with major historical events, such as the Great Recession or the COVID-19 pandemic, which caused the economy to behave in unusual ways that the models could not fully predict.

We also investigated whether changing the shape of the model through nonlinear transformations would make our predictions more accurate. In the first era (1980–2008), the relationship looked like a straight line, so no changes were needed. For the second era

(2009–2025), we tried using "log" and "square root" transformations to see if a curved line would fit the data better. However, these changes actually made the model less accurate by lowering the R^2 value, so we decided to stick with the original linear models for both time periods.

This study shows that the link between unemployment and labor force participation changed over time. From 1980 to 2008, the data supports the discouraged worker effect, as participation fell when unemployment rose. However, from 2009 to 2025, the relationship flipped, suggesting the added worker effect or other economic shifts. Ultimately, this proves that one single rule cannot predict worker behavior over 40 years, as major crises and historical changes constantly reshape the pattern.